



Effect of seagrass cover loss on seawater carbonate chemistry: Implications for the potential of seagrass meadows to mitigate ocean acidification

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ABSTRACT

Seagrass meadows are important marine ecosystems for mitigating ocean acidification because of their ability to raise the pH of seawater during the day. This ability may decrease as a result of the loss of these meadows, which is primarily caused by human activities and climate change. Here, we test the effect of seagrass cover loss on seawater carbonate chemistry to understand how the loss of seagrass meadows affects their ability to mitigate ocean acidification. pH, dissolved inorganic carbon (DIC), partial pressure of carbon dioxide (pCO₂), and aragonite saturation state (ΩAr) were measured in experimental tidal pools with varying proportions of seagrass coverage: 0% (mimicking a complete loss of seagrass meadows); 1%–29% (mimicking the greatest loss of seagrass meadows); 30%–59% (mimicking a moderate loss of seagrass meadows); and 60%–100% (mimicking the lowest loss of seagrass meadows). It was found that as seagrass cover decreased, pH and ΩAr levels in seawater decreased proportionally during the day, while pCO₂ and DIC increased. Additionally, correlation analysis showed a strong significant positive correlation between the seagrass cover and pH ($r_s = 0.9096$, $p < 0.0001$) and ΩAr ($r_s = 0.9031$, $p < 0.0001$), as well as a strong significant negative correlation between the seagrass cover and pCO₂ ($r_s = -0.9068$, $p < 0.0001$) and DIC ($r_s = -0.8947$, $p < 0.0001$). These results imply that the 7% annual global loss in seagrass meadows may limit seagrass meadows' ability to raise the pH of their surrounding seawater during the day, reducing their potential to mitigate ocean acidification. The study recommends that management strategies that minimize anthropogenic activities that cause seagrass loss be implemented in order for seagrass meadows to continue mitigating ocean acidification within their ecosystem and nearby ecosystems.

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1. Introduction

Globally, the concentration of atmospheric carbon dioxide (CO₂) is rapidly increasing, reaching levels of ~420 ppm in 2022, up from ~280 ppm before the industrial era (see <https://keelingcurve.ucsd.edu/>). The primary causes of this increase are the use of fossil fuels, the manufacturing of cement, and deforestation (Doney et al., 2009; Feely et al., 2009; Doney, 2006). It is estimated that the world's oceans have absorbed approximately 48% of emitted CO₂ since the industrial era (Sabine et al., 2004), and they continue to absorb approximately 30% annually (Barker and Ridgwell, 2012). In seawater, absorbed CO₂ undergoes a series of chemical reactions that result in a decrease in pH and aragonite saturation state (ΩAr), as well as an increase

in dissolved inorganic carbon (DIC) and partial pressure of CO₂ (pCO₂), a phenomenon known as ocean acidification (Jiang et al., 2019; Sabine et al., 2004).

Ocean acidification has received strong attention in scientific communities in recent years due to its negative impact on calcifying marine organisms such as corals and oysters, which are ecologically and economically important (Doney et al., 2020, 2009; Feely et al., 2009; Sabine et al., 2004). The calcification rate of calcifying organisms that precipitate carbonate ions for their growth, reef formation, and shell formation has been shown to be significantly reduced by ocean acidification (Doney et al., 2020, 2009; Doney, 2006). Coral reefs, for example, are expected to dissolve under future ocean acidification conditions, jeopardizing their ecological and socioeconomic importance (Albright et al., 2015).

Seagrass meadows are considered key marine ecosystems when it comes to ocean acidification mitigation due to their ability to raise pH and ΩAr while decreasing DIC and pCO₂ in

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Table 1

Species of seagrass observed in the experimental tidal pools. ✓ = present, X = absent, HSM = lowest loss of seagrass meadows (60–100% seagrass cover), MGM = moderate loss of seagrass meadows (30–59% seagrass cover), DGM = greatest loss of seagrass meadows (1–29% seagrass cover), and NSG = complete loss of seagrass meadows (0% seagrass cover).

Seagrass species	HSM	MGM	DGM	NSG
<i>Thalassia hemprichii</i>	✓	✓	✓	X
<i>Cymodocea rotundata</i>	✓	✓	✓	X
<i>Halodule</i> sp.	✓	✓	✓	X
<i>Halophila ovalis</i>	✓	✓	✓	X
<i>Cymodocea serrulata</i>	✓	X	X	X
<i>Zostera capensis</i>	X	✓	✓	X
<i>Thalassodendron ciliatum</i>	X	X	✓	X

their waters during the day (Ricart et al., 2021b; Bergstrom et al., 2019; Hendriks et al., 2014; Unsworth et al., 2012). This pH increase may help in mitigating ocean acidification within their ecosystem as well as nearby ecosystems (Hendriks et al., 2014; Unsworth et al., 2012). It was shown by Ricart et al. (2021a) and Semesi et al. (2009a) that calcifying organisms calcify best when coexisting with seagrass. However, the ability of seagrass meadows to mitigate ocean acidification is dependent on their coverage and may decrease if cover is lost (Nagelkerken and Connell, 2022). Seagrass meadows have declined by 29% globally and continue to decline at a rate of 7% per year (Waycott et al., 2009), owing primarily to human activities and climate change (Lugendo, 2016; George et al., 2018; Orth et al., 2006). However, it is unknown how the loss of seagrass meadows will affect their capacity to mitigate ocean acidification. As a result, there is a critical need to understand the effect of seagrass cover loss on seawater carbonate chemistry and its implications for seagrass meadows' ability to mitigate ocean acidification, as this is critical for supporting evidence-based policy making in seagrass meadow conservation. The objective of the study was to understand how the loss of seagrass meadows affects their ability to mitigate ocean acidification by examining the influence of seagrass cover loss on seawater carbonate chemistry using tidal pools in seagrass meadows that are accessible during low spring tides. We tested the hypothesis that the loss of seagrass meadows will result in lower pH and Ω_{Ar} conditions in their seawater during the day while increasing DIC and pCO_2 , limiting the ocean acidification mitigation potential of seagrass meadows.

2. Material and methods

2.1. Study site

This study was conducted in tidal pools within a seagrass meadow at Puna Beach, which is located at Lat. -7.0782 and Long. 39.54342 in south of Dar es Salaam, Tanzania. The tidal cycle is semidiurnal at the study site, with two high tides and two low tides during the day–night cycle. The intertidal area at Puna Beach has mixed-species seagrass meadows, with seven species found in the tidal pools used in this study (Table 1). Though not pristine, seagrass meadows at this site are subject to relatively low human pressures. Intertidal seagrass meadows at the study site are accessible and exposed to sunlight, hot air, strong insolation, and high evaporation rates during low spring tides (George et al., 2018). The tidal pools in the seagrass meadow studied are separated by at least 10 m, so that no water flows from one pool to another during low spring tides, mimicking a natural mesocosm and providing a controlled environment until the incoming high tide (Fig. 1).

Table 2

The depth and the area of tidal pools ($n = 10$).

	Depth (cm)	Area (m^2)
Mean	16.6	0.66
Standard deviation	4.73	0.18
Minimum	10.57	0.43
Maximum	23.67	1.05

2.2. Study design and data collection

Sampling was conducted in 40 experimental tidal pools with varying proportions of seagrass coverage, which were accessible during low spring tides. The experimental tidal pools were chosen using a purposive sampling strategy to ensure that they had the desired percentage of seagrass cover, no direct freshwater input, and no macrofauna such as sea urchins and sea cucumbers that may potentially influence the carbonate chemistry of seawater. The desired percentage of seagrass cover was delineated into four categories: 0% mimicking a complete loss of seagrass meadows (NSG) as well as an experimental control, 1%–30% mimicking the greatest loss of seagrass meadows (DGM), 31%–59% mimicking a moderate loss of seagrass meadows (MGM), and 60%–100% mimicking a lowest loss of seagrass meadows (HSM). The desired seagrass percentage cover in the experimental tidal pools was determined using a $0.5\text{ m} \times 0.5\text{ m}$ 0.25 m^2 quadrat. The pH, temperature, and salinity of the experimental tidal pools were measured between 11:00 a.m. and 2:00 p.m., when the area's seagrass meadows receive the most insolation and their productivity is likely to be at its peak (George et al., 2018). Sampling was done for three consecutive days from January 4th to 6th, January 2022, when the region's intertidal seagrasses often endure stable weather with high average temperatures (George et al., 2018, 2020).

2.3. Estimation of area and depth of the experimental tidal pools

Ten tidal pools were used to determine the average depth and area of experimental tidal pools (Table 2). The three points were measured in each pool using a ruler from the shallow point to the deeper part of the tidal pools and their average was used to represent the mean ($\pm$ standard deviation) depth of the experimental tidal pools. The mean ($\pm$ standard deviation) area of the pool was estimated by using the grid method, which determines the area of irregular shapes (Avril et al., 2004). The 0.25 m^2 quadrat used to estimate the percentage cover of seagrasses was divided into 100 squares, each containing 0.0025 m^2 . After the graduated quadrat was set on the surface of the pools and the number of squares counted as either full or half squares afterward, the surface area of the tidal pools was determined as follows:

$$\text{Area (m}^2\text{)} = (HS \times 0.5 \times 0.0025\text{ m}^2) + (FS \times 0.0025\text{ m}^2)$$

where FS is total number of full squares, and HS is total number of half squares obtained during counting. The mean depth and area of the experimental tidal pools were $16.6 \pm 4.73\text{ cm}$ and $0.66 \pm 0.18\text{ m}^2$, respectively (Table 2).

2.4. Measurement of pH, temperature, and salinity

pH and temperature were measured by using an ST20 OHAUS pH meter calibrated with standard buffers bought with the ST20 OHAUS pH meter (three-point calibration; 4.01, 7.01, and 10.01), while salinity was measured by using a handheld refractometer (PCE-010).

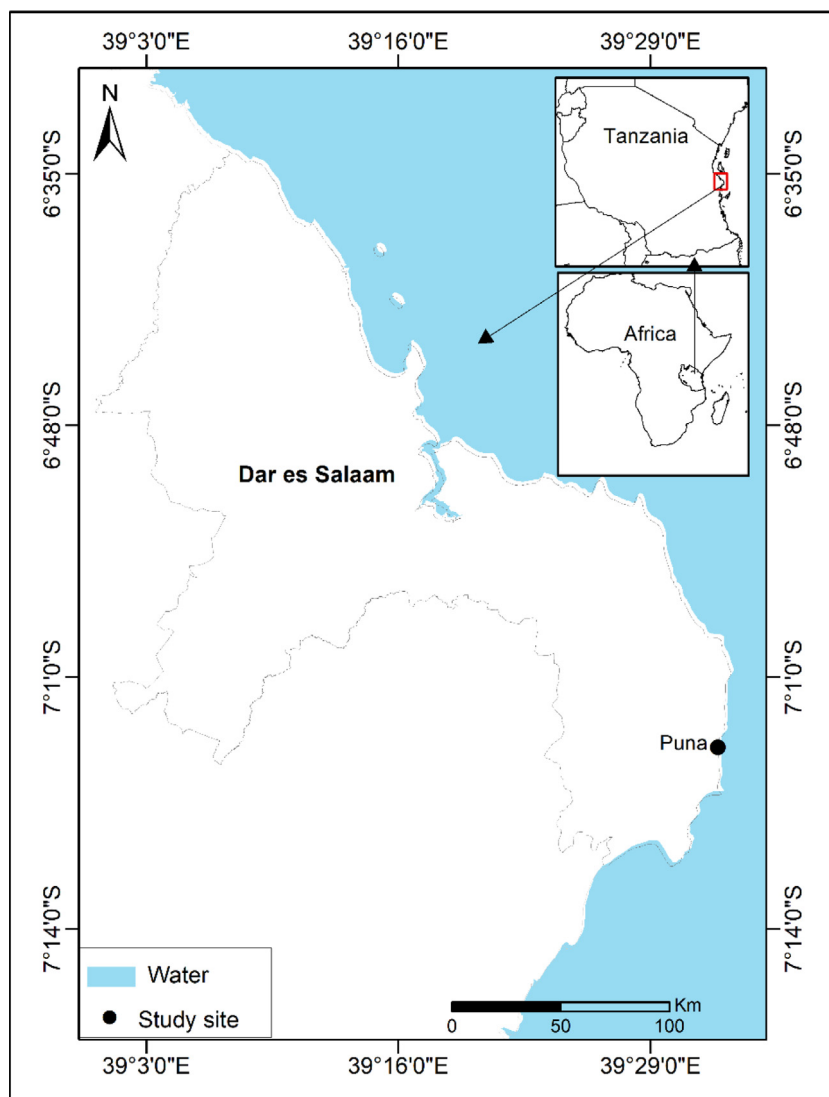


Fig. 1. A map showing the location of the study site.

2.5. Estimation of $p\text{CO}_2$, DIC, and ΩAr

According to Orr et al. (2018), when any two carbonate chemistry system parameters are present, the other two parameters can be determined. The average ocean alkalinity ($2300 \mu\text{mol/kg}$) and pH values measured from the experimental groups were used as carbonate species inputs together with salinity and temperature values measured from the experimental groups to calculate the remaining carbonate system parameters (DIC and $p\text{CO}_2$) using the CO_2 calculator Pelletier CO_2sys excel version (Pelletier et al., 2007). The chosen K_1 and K_2 constants were from Mehrbach et al. (1973), revised by Dickson and Millero (1987), while borate constants from Uppstrom (1974) were used. The total alkalinity (TA) in the experimental tidal pools can fluctuate due to seagrass productivity during the experiment (Semesi et al., 2009a,b), and the authors admit this possibility and its potential impact on the study's results. However, it is advised to use the average TA when predicting other carbonate chemistry system parameters when it is difficult to precisely estimate the TA values and their variability (Khan et al., 2020).

2.6. Data analysis

Prior to analysis, data for pH, $p\text{CO}_2$, DIC, and ΩAr were tested for normality using the Shapiro–Wilk test. However, all the data

failed the normality test. Therefore, the Kruskal–Wallis test (χ^2) was used to test for significant differences in pH, ΩAr , DIC, and $p\text{CO}_2$ among the experimental groups, followed by a Dunn's post-hoc test (Dunn, 1961). The Spearman ranking correlation coefficient was used to assess the relationship between seagrass cover and carbonate chemistry system parameters (pH, ΩAr , DIC, and $p\text{CO}_2$). All statistical tests were carried out in GraphPad Prism 8 at the 95% significance level.

3. Results

3.1. Physicochemical variables in the experimental tidal pools

Across all experimental tidal pools, water temperature ranged between 34.6 and 38.5°C and salinity between 33 and 35 (Table 3).

3.2. Effect of seagrass cover loss on pH, $p\text{CO}_2$, DIC, and ΩAr

The mean ($\pm$ SE) pH values in tidal pools with 60%–100% seagrass cover were highest (8.78 ± 0.1), followed by 30%–59% seagrass cover (8.66 ± 0.1) and 1%–29% seagrass cover (8.53 ± 0.08) and lowest in 0% seagrass cover (8.03 ± 0.17) (Fig. 2a; Table 4; $\chi^2 = 99.61$, $p < 0.0001$). Similar to pH, the mean ($\pm$ SE) ΩAr

Table 3

Summary statistics of temperature and salinity recorded in experimental tidal pools. HSM = lowest loss of seagrass meadows (60–100% seagrass cover), MGM = moderate loss of seagrass meadows (30–59% seagrass cover), DGM = greatest loss of seagrass meadows (1–29% seagrass cover), NSG = complete loss of seagrass meadows (0% seagrass cover), Min = Minimum, Max = Maximum, and SD = Standard deviation.

Variable	HSM	MGM	DGM	NSG
	Mean $\pm$ SD (Min–Max)	Mean $\pm$ SD (Min–Max)	Mean $\pm$ SD (Min–Max)	Mean $\pm$ SD (Min–Max)
Salinity	34.55 $\pm$ 0.69 (33–35)	34.69 $\pm$ 0.48 (34–35)	34.9 $\pm$ 0.32 (34–35)	34.80 $\pm$ 0.42 (34–35)
Temperature	35.97 $\pm$ 0.59 (35.1–37)	36.61 $\pm$ 0.83 (34.6–37.4)	36.42 $\pm$ 0.45 (35.8–37.2)	36.15 $\pm$ 0.91 (35–38.5)

Table 4

Summary statistics of pH, pCO₂, Ω Ar, and DIC across the experimental tidal pools. HSM = lowest loss of seagrass meadows (60–100% seagrass cover), MGM = moderate loss of seagrass meadows (30–59% seagrass cover), DGM = greatest loss of seagrass meadows (1–29% seagrass cover), NSG = complete loss of seagrass meadows (0% seagrass cover), Min = Minimum, Max = Maximum, and SD = Standard deviation.

Carbonate chemistry system parameters	HSM	MGM	DGM	NSG
	Mean $\pm$ SD (Min–Max)	Mean $\pm$ SD (Min–Max)	Mean $\pm$ SD (Min–Max)	Mean $\pm$ SD (Min–Max)
pH	8.78 $\pm$ 0.1 (8.66–9.00)	8.66 $\pm$ 0.1 (8.49–8.85)	8.53 $\pm$ 0.08 (8.38–8.64)	8.03 $\pm$ 0.17 (7.74–8.26)
DIC (μ mol/kg)	1304.12 $\pm$ 88.24 (1117.53–1406.22)	1386.98 $\pm$ 86.08 (1239.21–1538.98)	1502.34 $\pm$ 69.18 (1410.17–1624.81)	1900.29 $\pm$ 117.89 (1731.44–2086.12)
pCO ₂ (μ mol/kg)	31.28 $\pm$ 11 (11.22–47.66)	48.60 $\pm$ 18.96 (21.69–88.66)	79.72 $\pm$ 25.17 (60.31–128.94)	449.50 $\pm$ 241.56 (194.31–917.94)
Ω Ar	10.66 $\pm$ 0.58 (9.93–11.82)	9.99 $\pm$ 0.74 (8.69–11.15)	8.90 $\pm$ 0.69 (7.68–9.75)	4.75 $\pm$ 1.26 (2.78–6.54)

Table 5

The results of the Dunn's post hoc test between experimental groups. HSM = lowest loss of seagrass meadows (60–100% seagrass cover), MGM = moderate loss of seagrass meadows (30–59% seagrass cover), DGM = greatest loss of seagrass meadows (1–29% seagrass cover), NSG = complete loss of seagrass meadows (0% seagrass cover), Min = Minimum, Max = Maximum, and SD = Standard deviation.

Experimental groups	Seawater carbonate chemistry variables			
	pH	pCO ₂ (ppm)	DIC (μ mol/kg)	Ω Ar
HSM vs MGM	0.0410	0.0521	0.1761	0.2407
HSM vs DGM	<0.0001	<0.000	<0.0001	<0.0001
HSM vs NSG	<0.0001	<0.0001	<0.0001	<0.000
MGM vs DGM	0.0159	0.0149	0.0070	0.0017
MGM vs NSG	<0.0001	<0.0001	<0.0001	<0.0001
DGM vs NSG	0.0010	0.0009	0.0009	0.002

values were highest in tidal pools with 60%–100% seagrass cover (10.66 $\pm$ 0.58), followed by 30%–59% seagrass cover (9.99 $\pm$ 0.74) and 1%–29% seagrass cover (8.53 $\pm$ 0.08) and lowest in 0% seagrass cover (8.90 $\pm$ 0.69) (Fig. 2c; Table 4; $\chi^2 = 100.1$, $p < 0.0001$). Unlike pH and Ω Ar, the mean ($\pm$ SE) pCO₂ values were highest in tidal pools with 0% seagrass cover (449.50 $\pm$ 241.56), followed 1%–29% seagrass cover (79.72 $\pm$ 25.17) and in 30%–59% seagrass cover (48.60 $\pm$ 18.96) and lowest in 60%–100% seagrass cover (31.28 $\pm$ 11) (Fig. 2b; Table 4; $\chi^2 = 98.92$, $p < 0.0001$). Similar to pCO₂, the mean DIC values were highest in tidal pools with 0% seagrass cover (1900.29 $\pm$ 117.89), followed by 1%–29% seagrass cover (1502.34 $\pm$ 69.18) and 30%–59% seagrass cover (1386.98 $\pm$ 86.08), and the lowest in 60%–100% seagrass cover (1304.12 $\pm$ 88.24) (Fig. 2d; Table 4; $\chi^2 = 96.78$, $p < 0.0001$). Dunn's test further revealed that there was a significant difference in pH, Ω Ar, DIC and pCO₂ levels between experimental groups (Table 5). Only pH levels varied significantly between experimental groups with varying proportions of seagrass cover. Ω Ar, DIC and pCO₂ levels varied significantly between experimental groups with varying proportions of seagrass cover, with the exception of between 30%–59% seagrass cover and 60%–100%

seagrass cover experimental groups. Significant differences in pH, Ω Ar, DIC, and pCO₂ were found between experimental groups with seagrass cover and those without (Table 5).

3.3. Correlation between seagrass cover and pH, pCO₂, DIC and Ω Ar

While the seagrass cover had a significantly positive correlation with pH ($r_s = 0.9096$, $p < 0.0001$) and Ω Ar ($r_s = 0.9031$, $p < 0.0001$), it had a significantly negative correlation with pCO₂ ($r_s = -0.9068$, $p < 0.0001$) and DIC ($r_s = -0.8947$, $p < 0.0001$) (Fig. 3) indicating that as the seagrass cover decrease the pH and Ω Ar decrease also while pCO₂ and DIC increases.

4. Discussion

This study shows that the loss of seagrass meadows due to anthropogenic activities and climate change causes a proportional decrease in pH and Ω Ar levels in their seawater during the day, as well as an increase in pCO₂ and DIC. Consequently, this would limit the ability of seagrass meadows to mitigate ocean acidification in both their own ecosystem and nearby ecosystems.

The productivity of seagrass influences the carbonate chemistry of seawater by lowering pCO₂ and DIC while raising pH and Ω Ar during the day (Hendriks et al., 2014; Semesi et al., 2009a,b). The loss of seagrass cover may have resulted in a decrease in productivity (Ricart et al., 2021b; Cyronak et al., 2018), which could have accounted for the observed proportional increase in pCO₂ and DIC, as well as a decrease in pH and Ω Ar during the day. These findings indicate that seagrass loss would limit the potential of seagrass meadows to mitigate ocean acidification, both within and in nearby ecosystems. To the best of our knowledge, this is the first study to report on the impact of seagrass cover loss on the capacity of seagrass meadows to mitigate ocean acidification.

Seagrass photosynthesis and respiration are key processes that regulate the levels of pH, Ω Ar, DIC and pCO₂ in seawater (Hendriks et al., 2014; Semesi et al., 2009a,b). The photosynthetic

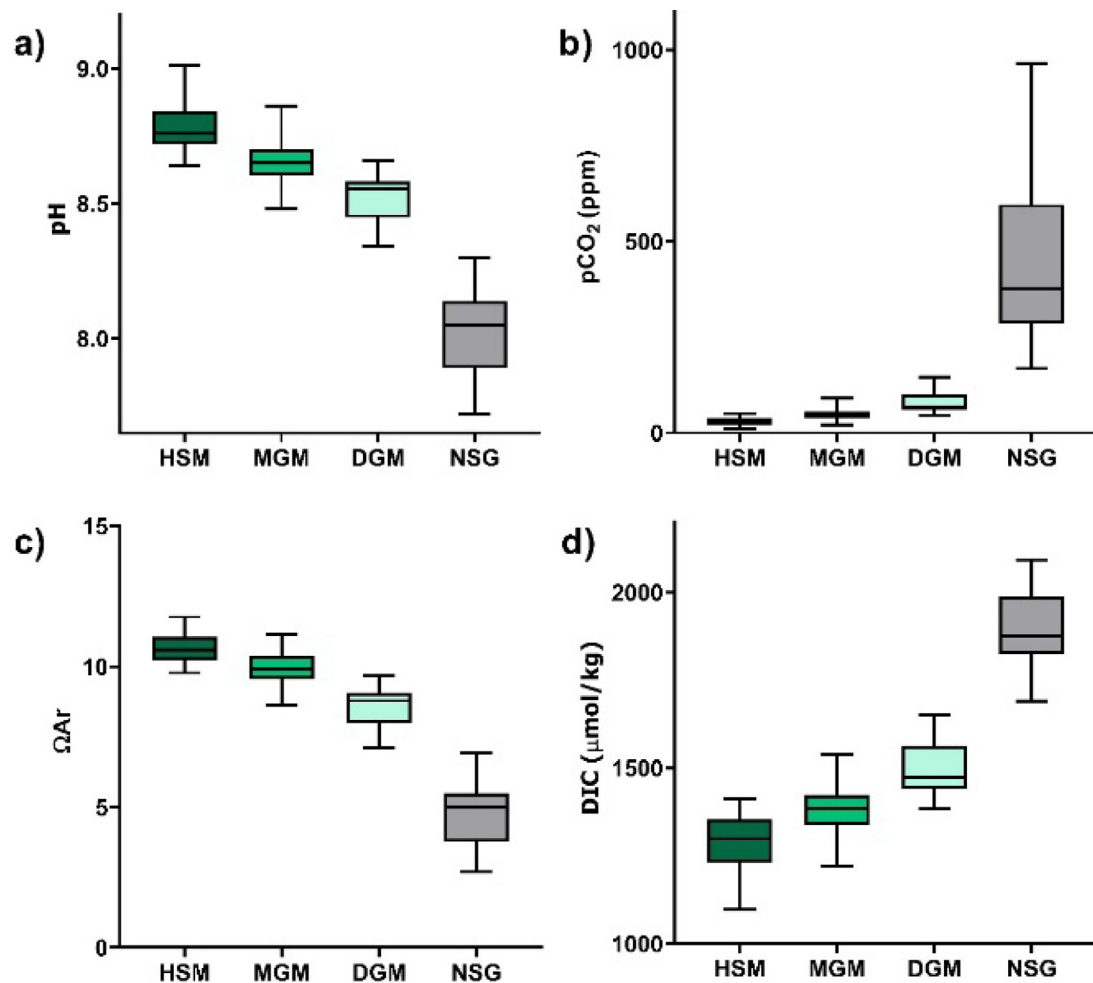


Fig. 2. Mean $\pm$ SE of pH (a), pCO₂ (b), ΩAr (c) and DIC (d) among experimental tidal pools. HSM = lowest loss of seagrass meadows (60%–100% seagrass cover), MGM = moderate loss of seagrass meadows (30%–59% seagrass cover), DGM = greatest loss of seagrass meadows (1%–29% seagrass cover), and NSG = complete loss of seagrass meadows (0% seagrass cover).

process of seagrass occurring during daytime consumes CO₂ to produce organic carbon, which causes an increase in the pH and ΩAr while decreasing the DIC and pCO₂ of seawater (Unsworth et al., 2012; Tang and Hadibarata, 2022; Ricart et al., 2021b; Hendriks et al., 2014). The respiration process of seagrass, on the other hand, causes an increase in DIC and pCO₂ in seawater while lowering pH and ΩAr (Hendriks et al., 2014). The dominance of seagrass photosynthesis or respiration processes determines their positive effects on pH, ΩAr, DIC, and pCO₂ in a system (Van Dam et al., 2021). During the day, photosynthesis outweighs respiration in healthy seagrass meadows, resulting in high pH and ΩAr levels in the seawater (Ricart et al., 2021b; Bergstrom et al., 2019; Hendriks et al., 2014; Unsworth et al., 2012). In this study, the pH increased up to 8.78 during the day at low spring tides in tidal pools with a highest proportion of seagrass cover (60–100%), which is comparable to the findings of Duarte and Krause-Jensen (2018), who found an increase in pH of up to 9.0 in seagrass tidal pools, and Semesí et al. (2009a), who found a pH value of 8.9. However, as the proportion of seagrass cover decreased, the pH values decreased, demonstrating the negative effect of seagrass cover loss on seagrass meadows' ability to mitigate ocean acidification. This indicates that tidal pools with high proportions of seagrass cover can serve as ocean acidification refuges during the day. These findings concur with those of Kwiatkowski et al. (2016), who predicted that the tidal pools

with seagrass along the Northern California coast will serve as refuge from ocean acidification. The tidal pools without seagrass, on the other hand, had the lowest pH and ΩAr as well as the highest pCO₂ and DIC, indicating that they are ocean acidification hotspots during daytime in spring low tides and that coastal areas without seagrasses are more likely to be impacted by future ocean acidification conditions.

Anthropogenic activities and climate change reduce the productivity of tropical seagrasses, resulting in low shoot density and biomass (George et al., 2018). The strong positive correlation observed in this study between seagrass cover and pH, and ΩAr suggests that disturbances that result in a seagrass cover decline would reduce the capacity of seagrass meadows to mitigate ocean acidification. The effect of disturbance on productivity of seagrasses have been shown to result in net heterotrophic conditions in seagrass meadows with higher respiration than photosynthesis, potentially amplifying ocean acidification (Van Dam et al., 2021, 2019). This finding is supported by Duarte et al. (2010), who predicted that future anthropogenic activities and climate change would shift seagrasses meadows from net autotrophic to net heterotrophic ecosystems. The negative significant relationship between seagrass cover and pCO₂ and DIC may indicate the effectiveness of seagrass in removing CO₂ from seawater via photosynthesis. The productivity and biomass of seagrass have been demonstrated to increase at higher CO₂ concentrations,

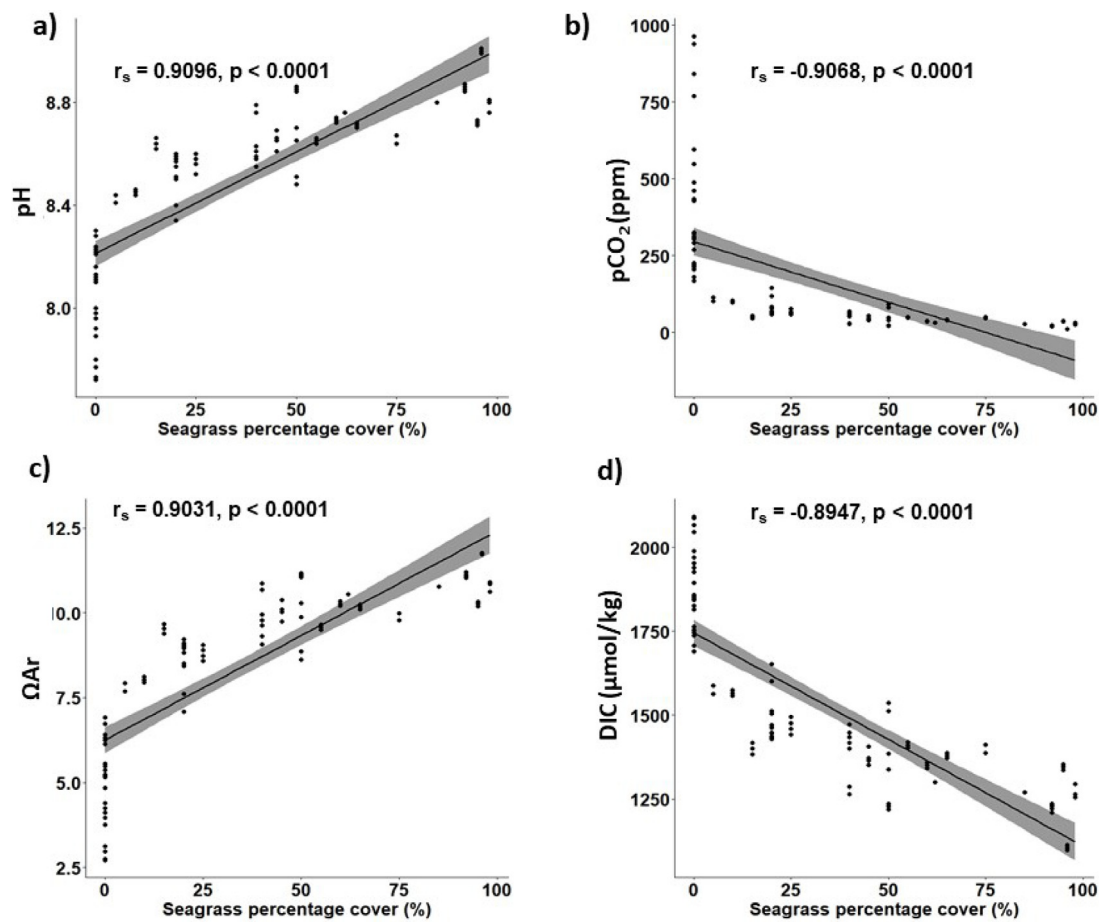


Fig. 3. The correlation between the seagrass percentage cover and pH (a), pCO_2 (b), ΩAr (c), and DIC (d).

such as those found in volcanic CO_2 seeps in New Papua Guinea (Russell et al., 2013), suggesting that seagrass will benefit from future ocean acidification conditions characterized by high CO_2 concentrations. Considering these findings, it is imperative that management actions aiming at minimizing anthropogenic activities that reduce seagrass cover should be implemented to allow seagrass meadows to continue mitigating ocean acidification.

5. Conclusion

Based on the results of this study, anthropogenic- and climate change-induced losses of seagrass meadows result in low pH and ΩAr levels in their seawater during the day while increasing pCO_2 and DIC levels. This is likely to reduce the capacity of seagrass meadows to mitigate ocean acidification in their ecosystem and nearby ecosystems such as mangroves and coral reefs. Protecting, conserving, and restoring seagrass meadows with the goal of increasing their coverage is highly recommended to improve their capacity to mitigate ocean acidification.

CRediT authorship contribution statement

Samson Job: Conceptualization, Methodology, Data collection, Investigation, Data curation, Writing – original draft, Writing – review & editing. **Baraka Sekadende:** Writing – review & editing. **Gloria Yona:** Writing – review & editing. **Rushingisha George:** Conceptualization, Methodology, Supervision, Writing – review & editing. **Blandina Robert Lugendo:** Conceptualization, Methodology, Supervision, Writing – review & editing. **Ismael Aaron Kimirei:** Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships that may be considered as potential competing interests: Samson Job reports financial support was provided by Western Indian Ocean Marine Science Association (WIOMSA). Ismael Kimirei reports a relationship with WIOMSA that includes: funding grants.

Data availability

Data will be made available on request.

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